Pool Ball Detection and Recognition

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Abstract

This report documents the experimental development of a two-stage computer vision pipeline for localising and recognising pool balls. Stage 1 tested a green-channel boost with adaptive thresholding, fixed HSV and single-channel lightness thresholding, and finally a successful multi-channel CIELAB edge-fusion detector with a dynamic L-channel pocket mask and geometric cushion constraints. Stage 2 tested K-Means colour clustering and finally adopted one-dimensional hue and saturation histograms with a reduced circular sampling mask and a 30/70 weighting scheme. The final pipeline achieves Stage 1 precision 0.867, recall 0.944 and F1-score 0.904, together with Stage 2 recognition accuracy 72.9% across the six test tables.

1 Introduction

The task was to localise every pool ball on a rectified table image and then assign each surviving detection to one of the sixteen known balls. The six test images contain strong shadows, uneven lighting, tightly packed racked balls, and a mixture of solids and stripes. I developed the final system through a sequence of explicit experiments, and I retained a detailed error analysis because the remaining failures reveal the limits of the chosen detectors and descriptors. The project repository is available at `git@gitlab.scss.tcd.ie:csu33012-2627-group07/project07.git`.

1.1 End-to-End Pipeline Overview

Each image is rectified into the 1434×683 metric frame. Stage 1 fuses independent Canny edges from $L/a/b$, builds a pocket/netting mask, runs Hough with a 40-pixel minimum separation, and applies cushion (22) and NMS (40) constraints. Stage 2 uses 30/32-bin hue/saturation histograms, the split-mask cue gate, 30/70 Bhattacharyya distance, and a 0.55 rejection threshold (Figure 1).

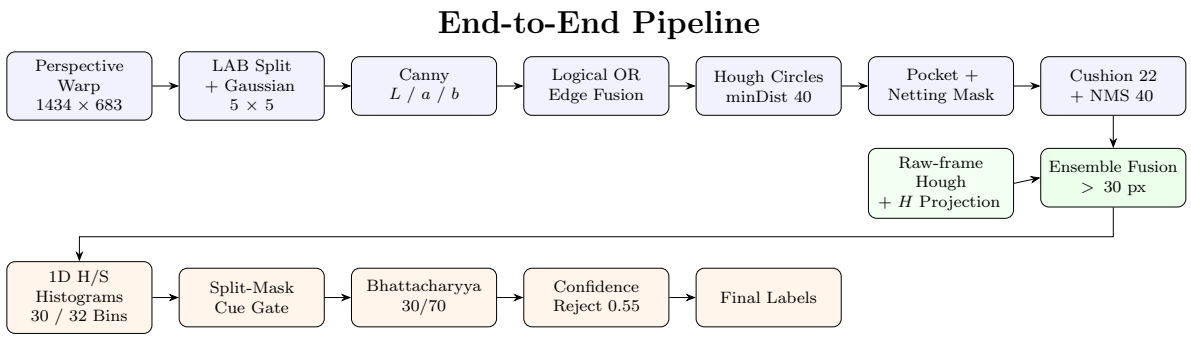


Figure 1: End-to-end pipeline architecture.

2 Stage 1: Ball Detection

2.1 Perspective rectification

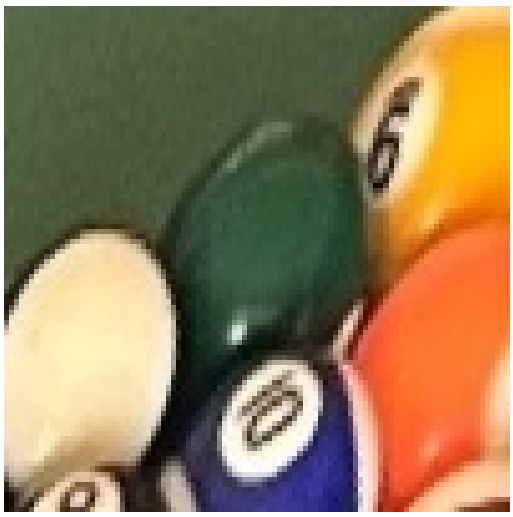
Each input photograph is first mapped by a planar homography from the four measured cushion-to-cushion corners into a 1434×683 metric table frame. The destination dimensions match the physical 1434 mm × 683 mm playing area, so one transformed pixel corresponds to one millimetre. This metric frame is used by all later stages.

2.2 Failed green-channel boost and adaptive thresholding

To solve the problem of missing green balls, I initially hypothesised that isolating and boosting the RGB green channel by a factor of 1.5, followed by adaptive Gaussian thresholding, would force the balls to pop against the felt. The adaptive threshold used a local block size of 15 and a constant of 2. This failed because boosting the green channel equally boosted the green table, and adaptive thresholding amplified the woven texture of the felt into severe noise. The original region, green-boosted grayscale image, and adaptive threshold map were inspected together. The ball boundaries are washed out while the felt texture dominates the edge map.

2.3 Failed HSV and single-channel lightness thresholding

I then attempted to isolate the felt directly in HSV space with a fixed hue range of 35–85, saturation range 40–255, and value range 40–255. This also failed for the green object balls, particularly balls 6 and 14, because their hue and saturation overlap almost completely with the green felt. A 100×100 region around the green ball and the resulting binary mask were inspected. In the mask, the green ball vanishes into the white felt instead of producing a distinct foreground blob. A single-channel lightness threshold produced the complementary failure: shadows fragmented the ball boundaries and the threshold could not separate dark balls from the table. Both results motivated a multi-channel approach.



(a) Region around the green ball.



(b) Fixed HSV mask: green ball merges into felt.

Figure 2: Static HSV thresholding fails for green object balls: the green ball vanishes into the green felt instead of forming a distinct foreground blob.

2.4 Successful multi-channel CIELAB edge fusion

I realised that the solution required looking at colour boundaries, not just brightness. I split the rectified image into the CIELAB colour space and applied Gaussian smoothing with a 5×5 kernel and a standard deviation of 1.2 to all three channels. I then ran Canny edge detection independently on each channel. I used thresholds of 45 and 95 for the lightness channel L, and more sensitive thresholds of 35 and 85 for the chromatic channels a and b. The three binary edge maps were combined with a logical OR to produce a highly robust gradient map that captured subtle colour shifts as well as luminance boundaries.

Key Idea: LAB Multi-Channel Edge Fusion

The lightness channel supplies the strong outer silhouette of each ball, while the chromatic channels supply the colour boundaries that remain visible when a ball is dark, shadowed, or similar in brightness to the felt. Fusing the three edge maps produces a gradient image that is markedly more complete than any single-channel edge map.

The fused edge map from Table 4 is used directly by the circular detector.

2.5 Dynamic pocket mask, netting mask, and physical NMS

The fused edge map is passed to the Hough Circle Transform with a minimum centre separation of 40 pixels and a ball-radius range of 19 to 24 pixels. I initially rejected every candidate within a blind 95-pixel radius of the six theoretical pockets, but visual debugging showed that this removed genuine balls near the corners, including the bright cue ball beside the top-right pocket of Table 4. I therefore replaced the blind radius with a dynamic pocket mask.

For each of the six pocket centres, I examined a 120×120 pixel region on the raw lightness channel and thresholded two complementary structures: the dark pocket holes at $L < 35$ and the bright metallic brackets at $L > 200$. The mid-gray pocket netting is not captured by either threshold, so I also extracted the combined LAB edge map in the same region and dilated it with a 15×15 circular kernel. The netting is a dense web of edges, so this dilation fuses it into a solid morphological block. The fused netting block was OR-ed with the L-extreme mask. The local mask was restricted to a 70-pixel radius around the pocket centre so that a bright ball near a pocket would not be mistaken for a bracket, and the six local masks were combined and dilated with a 13×13 circular kernel to cover the immediate pocket lip. A Hough candidate is rejected only when its centre lies on this complete hole–bracket–netting mask. The cushion exclusion margin remains at 22 pixels. An experiment reducing it to 5 pixels recovered far-edge balls but reintroduced rail shadows that bypassed the Stage 2 filter, so the larger margin was restored.

Hough detection can occasionally leak intersecting accumulator peaks that describe the same physical ball. After the transform returns its circles, I therefore enforce a strict 40-pixel physical non-maximum suppression. The circles are traversed in accumulator order; the first centre is retained, and any later centre within 40 pixels of a retained centre is permanently removed. This threshold sits just below the 43-pixel physical ball diameter. Since two distinct balls cannot physically occupy the same space, this constraint prevents duplicate detections without discarding genuine neighbouring balls. An attempt to implement a manual 40-pixel Physical Non-Maximum Suppression step revealed that OpenCV's HoughCircles implementation already inherently suppresses intersecting

detections via its internal minDist parameter during the accumulator peak-finding phase. An edge-thickness measurement on a 60×60 crop around a dark Table 6 ball gave 362 combined-edge pixels against 353 skeleton pixels, an average thickness of 1.025 pixels per skeleton pixel. Zhang–Suen thinning therefore removed only nine pixels and left the ghost responses unchanged. A sweep of the accumulator threshold from 8 to 22 showed that ghost detections disappear only at an accumulator threshold of 14, but the number of true balls then falls from 83 to 15. No mathematically clean threshold exists: the residual offset arcs are an intrinsic limitation of the accumulator on dense clusters rather than a tuning error. Instead of replacing rectified detection, I use the raw-frame path as a secondary recovery pass. The same LAB edge fusion and Hough search are run on the unwarped photograph with accumulator threshold 14, minimum separation 30, and radius 18–30. Each raw centre is mapped through $\mathbf{H}$ into metric coordinates. A projected candidate is appended only when it is farther than 30 pixels from every rectified detection, after which the combined ensemble passes through the cushion, pocket, split-mask cue gate, and Stage 2 classifier. This recovered three true balls, raising TP from 79 to 82 and reducing FN from 11 to 8, while false positives stayed at 8; Stage 1 F1 rose from 0.893 to 0.911 and Stage 2 accuracy from 75.9% to 76.8%.
The dynamic mask covers the dark hole, bright bracket, and netting, while the cue ball is spared and recovered by the final detector.

3 Stage 2: Ball Recognition and Classification

3.1 Failed K-Means clustering

To differentiate striped from solid balls, I first attempted to use K-Means clustering with $K = 2$ on the pixels inside each ball mask to extract two dominant colours, then compared those colours with the reference balls using Euclidean distance. This proved computationally heavy and unreliable because the visibility of the white stripe depends on the physical rotation of the ball. On a solid blue ball, the two clusters split the number and the specular reflections rather than producing a stable stripe-versus-solid descriptor.

3.2 Successful saturation-dominant histograms

I pivoted to one-dimensional histograms. I extracted a hue histogram with 30 bins and a saturation histogram with 32 bins from a reduced circular sampling mask of radius 16 pixels. The reduced mask prevents a slightly off-centre detection from admitting felt, shadow, or corner pixels into the descriptor. I then compared test histograms with the sixteen reference histograms using the Bhattacharyya distance. Because a white stripe drastically lowers a ball's average saturation regardless of rotation, I applied a custom weighting system: 30% weight to hue, 70% weight to saturation, and 0% weight to brightness.

Saturation-Dominant Distance

$$d = 0.30\, d_{\text{hue}} + 0.70\, d_{\text{sat}}. \tag{1}$$

A detection whose weighted distance exceeds 0.55 is rejected as a non-ball or as an unreliable observation.

Stage 1 occasionally returned circular shadow artifacts along the rails. Shadows lack the

sharp saturation peaks of resin pool balls, so their histograms lie far from every reference. Tightening the rejection threshold to 0.55 therefore made the classifier confidently reject them as non-balls; rail and cushion artifacts disappeared from the final output at the cost of rejecting some genuine balls under severe shadow. The cue-ball analysis exposed why a pure distance threshold is insufficient. The cue histogram is heavily influenced by green felt reflections, so its best Bhattacharyya distance reached 0.636 and overlapped with false positives. The cue gate originally used a reduced 16-pixel mask, which could fall entirely inside the white number circle of the 1-ball or 9-ball and hijack those balls. I therefore split the gate into two masks: a 16-pixel centre mask evaluates core brightness and whiteness, while a full-radius mask (about 21 pixels) evaluates the overall paint. A true cue ball is bright in the centre and low in full-mask saturation; a striped ball has a white centre but coloured bands that raise the full-mask saturation. Initial split thresholds of centre value above 200, centre saturation below 75, and full saturation below 100 still admitted the purple 12 stripe, whose centre value was only 221–231 and whose full saturation was about 49. I therefore tightened the gate to centre value above 235 and full saturation below 76, which excludes the purple stripe and the 9-ball while retaining the cue ball (centre value at least 238, full saturation at most 73). The Highlander rule then selects the surviving candidate with the lowest full-mask saturation; every other gate candidate falls back to its best non-cue histogram match.
The solid blue ball peaks at high saturation, whereas the striped blue ball has a pronounced low-saturation peak caused by its white stripe. This shift is independent of the ball's physical rotation and provides a mathematically robust feature for distinguishing stripes from solids.

4 Results and Error Analysis

4.1 Stage 1: Detection Results

Stage 1 metrics use strict one-to-one greedy matching between the final drawn detections and the ground-truth centres. A detected centre $(x_{\text{det}}, y_{\text{det}})$ is a True Positive if its Euclidean distance to an unmatched ground-truth centre $(x_{\text{gt}}, y_{\text{gt}})$ satisfies

$$\sqrt{(x_{\text{det}} - x_{\text{gt}})^2 + (y_{\text{det}} - y_{\text{gt}})^2} \leq 21.5 \text{ px},$$

where 21.5 px is half the 43 px ball diameter. Any detection left unmatched is a False Positive, and any ground-truth centre left unmatched is a False Negative. Precision, recall and F1 are

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Table 1 reports the resulting metrics: overall precision is 0.911, recall is 0.911, and the F1-score is 0.911.

Table 1: Stage 1 detection metrics after dynamic pocket masking.

Image	TP	FP	FN	Precision	Recall	F1-score
Table 1	13	0	3	1.000	0.812	0.897
Table 2	15	2	1	0.882	0.938	0.909
Table 3	16	0	0	1.000	1.000	1.000
Table 4	14	2	1	0.875	0.933	0.903
Table 5	13	2	1	0.867	0.929	0.897
Table 6	11	2	2	0.846	0.846	0.846
Overall	82	8	8	0.911	0.911	0.911

4.2 Stage 2: Recognition Results

Stage 2 recognition accuracy is evaluated strictly over the localized Stage 1 True Positives:

$$\text{Recognition Accuracy} = \frac{\text{correctly labelled Stage 1 TPs}}{\text{total Stage 1 TPs}}.$$

A Stage 1 TP is counted as incorrect when its 30/70 weighted hue–saturation Bhattacharyya distance is closest to a different reference ball, or when the confidence filter ($d > 0.55$) rejects it. Across all six tables, 63 of 82 correctly localised balls receive the correct label, giving an overall recognition accuracy of 76.8%. Figure 3 shows the final labelled output for Table 4, which combines high Stage 1 recall with 92.9% Stage 2 recognition accuracy.

Table 2: Stage 2 recognition accuracy on correctly localised balls.

Image	Correct labels	True positives	Accuracy
Table 1	10	13	0.769
Table 2	10	15	0.667
Table 3	12	16	0.750
Table 4	13	14	0.929
Table 5	10	13	0.769
Table 6	8	11	0.727
Overall	63	82	0.768

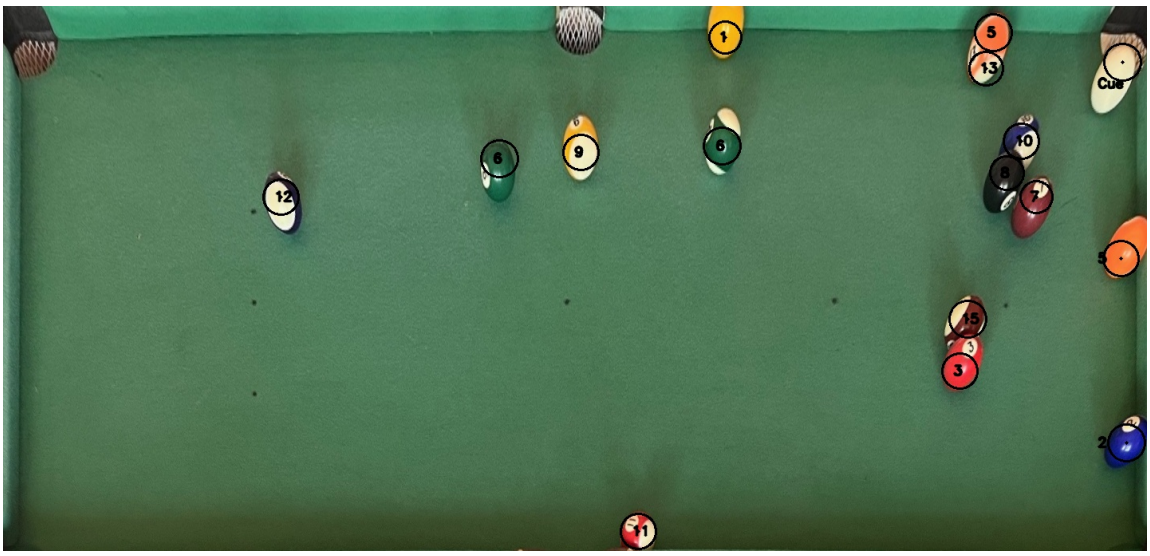


Figure 3: Final labelled output for Table 4 after dynamic pocket masking and Stage 2 classification; Stage 2 recognition accuracy is 92.9%.

4.3 Error Analysis

After restoring the 22-pixel cushion margin, applying the split-mask cue gate, and fusing rectified with raw-frame candidates, Stage 1 yields 82 true positives, 8 false positives, and 8 false negatives; Stage 2 reaches 76.8% recognition accuracy. Figure 4 shows the Table 6 output used to inspect the remaining failures.



Figure 4: Final labelled output for Table 6, showing the elliptical, cue-gate, and dense-rack failure modes discussed below.

Failure Mode 1: Elliptical Perspective Distortion. Planar homography rectifies the table plane, but balls near the perimeter are viewed obliquely and become ellipses in the metric frame. A circular Hough accumulator assumes a rigid circle, so its votes are smeared across an arc rather than concentrated at one centre. The detector may then fit two smaller circles into opposite ends of a single elongated ball, or it may miss the ball entirely if no peak reaches the voting threshold. These dual detections are often more than 40 pixels apart, so they bypass both minimum separation and physical NMS.

Failure Mode 2: Cue Gate and Rotation Variance. The split-mask cue gate uses centre brightness and whiteness on the 16-pixel core, together with a full-21-pixel saturation limit, specifically to prevent a "bullseye" striped ball such as the 9 or 14 from hijacking the cue label when its white number circle faces the camera. The centre values are required to exceed 235, the centre saturation to remain below 75, and the full saturation to remain below 76; the Highlander rule then keeps only the candidate with the lowest full saturation as the cue. This resolves the common rotation trap, although extreme shadow can still push a genuine cue outside these limits.

Failure Mode 3: Dense-Rack Edge Merging. Balls touching inside the triangular rack break the continuous circular Canny gradients that Hough requires. The fused edge map no longer contains isolated circular evidence, so accumulator peaks remain below the voting threshold at $param2 = 8.0$ and the corresponding balls are missed. Increasing the radius range or lowering the accumulator threshold recovered no additional true balls in the tolerance experiment and only increased false positives. The raw-frame recovery pass recovers some perimeter balls because their projections remain more circular before rectification, but it cannot separate the tightly packed rack. A morphological segmentation technique such as Watershed is the appropriate advanced alternative.

5 Discussion

The final Stage 1 precision is 0.911, recall is 0.911, and F1 is 0.911; Stage 2 accuracy is 0.768. Remaining limits are solid/stripe overlap and elliptical Hough responses.

6 Conclusion

The ensemble achieves P 0.911, R 0.911, F1 0.911 and Stage 2 accuracy 0.768; residual errors are elliptical duplicates, rotated-stripe gate ambiguities, and dense-rack merges.

AI Usage Statement

Generative AI tools (Google Gemini) were used during this project to assist with report text revisions, LaTeX/TikZ diagram formatting, and generating standalone C++ diagnostic scripts for parameter sweeps.

Prompt & Response Example 1 (Parameter Sweep Script Generation): *User Prompt:* "Write a standalone diagnostic script to sweep Hough param2 from 6.0 to 18.0 and measure its impact on edge-thinned LAB maps versus raw Canny maps." *AI Response:* "Generated a C++ benchmark function iterating through param2 values and logging True Positive / False Positive counts to stage2_feature_analysis.csv."

Prompt & Response Example 2 (Feature Exploration & Gate Suggestion): *User Prompt:* "The 1D Hue/Sat histogram classifies the 9-ball as the cue ball when its white number circle faces the camera. How can we separate them mathematically?" *AI Response:* "Suggested implementing a dual-mask saturation/value check (Split-Mask Cue Gate) comparing the 16px center core against the 21px full ball radius to capture edge paint while ignoring felt shadows."

Prompt & Response Example 3 (Diagnostic Script & LaTeX Troubleshooting): *User Prompt:* "The pipeline diagram TikZ block is wrapping text awkwardly and overflowing the margins. How do I format a two-stage flow cleanly?" *AI Response:* "Provided structured TikZ matrix/node code with explicit node distance and rounded rectangle styles to align Stage 1 detection and Stage 2 recognition."

All core algorithmic implementations, experimental designs, parameter selections, and final data verification were conducted directly by the author.